



Experiments and analysis of the internal wall pressure of a ducted rotor

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ARTICLE INFO

Article history:

Received 3 July 2018

Revised 21 December 2018

Accepted 22 December 2018

Available online 5 March 2019

Handling Editor: L. Huang

Keywords:

Fan acoustics

Duct acoustics

Duct Green's function

ABSTRACT

This article presents unsteady wall pressure measurements from a low-speed, low-solidity rotor operating in a circular duct. A circular microphone array was placed within the duct downstream of the rotor to record the unsteady surface pressure. These data were decomposed into spatial modes and then further decomposed into hydrodynamic and acoustic pressure fluctuations. The acoustic pressure was then further separated into the acoustic source function and an acoustic transfer function. The acoustic source function from the rotor compared well with a previous formulation for turbulence interaction noise. The transfer function showed a complicated relationship with frequency, with significant local maxima and minima. These features of the duct acoustics were studied using a simplified Green's function type approach, using the method of images to represent the duct end effects. The simple analytical method was able to capture many of the detailed features of the measurements with qualitative agreement.

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1. Introduction

Flow-induced sound and vibration is an important aspect of axial flow turbomachinery design. Rotor-relative flow speeds are often high enough to result in sound and vibration issues that are caused by rotating components. Rotating noise sources can be located in a short, annular duct which leads to a complicated interior pressure field. This, in turn, can lead to radiated sound and structural vibration that is challenging to predict.

The purpose of the present work was to study the interior acoustics of a simple, axisymmetric, ducted-rotor system. The objective was to first obtain experimental measurements of the unsteady wall pressure created by a low-speed rotor operating in a finite-length duct. The second objective was to study and understand these results. The latter task utilized a simple analytical model that provided qualitative understanding of the unique features and characteristics observed in the wall pressure data.

A great deal of literature exists that is focused on the acoustics within a duct. Doak [1] reviewed fundamental aspects of duct acoustic theory in the context of Lighthill's analogy. Nayfeh et al. [2] summarize methods of determining the parameters associated with acoustic radiation of the engine duct system of an aircraft. Abrahamsson and Kreiss [3] presented a fast-solving numerical solution to the coupled mode equations of duct acoustics. More recently, authors such as Kang and Kim [4] and Huang [5] have examined methods of controlling radiated duct noise.

Previous authors have examined various methods by which sound can be generated by a ducted rotor. Golubev and Atassi [6] provided an analytical method for predicting the propagation of acoustic waves in a duct with a swirling flow. They found that the resulting refraction caused different acoustic wave propagation speeds in the upstream and downstream direction.

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Glegg and Jochault [7] provided analytical predictions of fan self-noise on a high-solidity fan, correcting for the correlated surface pressures on different rotor blades. Lee et al. [8] presented an analytical model for predicting the acoustics generated by the vortex shedding of an axial-flow ducted rotor. The sound generated by rotor interactions with ingested flow irregularities were analytically examined by, for example, Atassi and Fang [9] and Kobayashi [10]. Further analysis of the sources of rotor acoustics can be found in Chapter 4 of the book by Goldstein [11] and the review papers by Morfey [12], Wright [13], and Huff [14].

The radiated sound generated by a rotating blade row can be considered to be a combination of both the “acoustic sources” and an “acoustic transfer function”. In many applications the acoustic source can be considered to be a dipole-like sound mechanism that results from the interaction of rotating blades with a turbulent approach flow [15]. The following will discuss the literature related to rotor acoustic source modeling in order to set the context for the present investigation. Hanson [16] studied turbulence ingestion noise generated by both periodic inflow disturbances and large scale anisotropic turbulence. The study identified sharp peaks in the resultant acoustic spectra from both sources. Moiseev et al. [17] studied a ducted rotor with various inflow conditions and numbers of rotor blades, concluding that most of the sound was generated due to the interaction of the hub and tip regions of the rotor with the ingested boundary layer. Ganz et al. [18] performed an experiment which involved the capability to remove the boundary layer from the ingested flow and concluded that the boundary layer was a significant source of sound in a ducted rotor. Martinez [19] developed a model for determining the thrust and ultimately acoustic radiation of a rotor subjected to anisotropic turbulence. The analysis provided a physical explanation for the broad humps in the pressure-frequency spectra, deemed the “haystacks”, around multiples of the blade passing frequency. Atassi and Logue [20] predicted broad humps at multiples of the blade passing frequencies by using rapid distortion theory to develop a wavenumber spectra model that can account for anisotropic turbulence.

The present work will build upon the initial results of Stephens and Morris [21], where a force-transfer function separation algorithm was used to experimentally determine the acoustic forcing function and acoustic transfer function of a ducted rotor for a single observer external to the duct. In a second study, Stephens and Morris [22] provided an analytical method for utilizing the statistics of the approach turbulence to predict the radiated sound from a ducted rotor. The method approximated the net unsteady force of the blades as a single, compact dipole acoustic source. The predictions were shown to match well with measurements of the radiated sound, but because the acoustics were approximated by a single, on-axis dipole, the model could not predict circumferential dependence of the acoustics within the duct. Note that the spatial distribution of the acoustic pressure can be important because the structural response of a duct depends on both the magnitude of the wall pressure and on its spatial characteristics.

The following section describes the ducted rotor experiment, the boundary conditions, and the measurement methods. The boundary conditions were designed to be simple enough to be analytically tractable, yet representative of a level of complexity that exists in more realistic engineering configurations. The resulting data set is unique in the current literature, and hence it provides new insight into qualitative and quantitative features of the problem. The results may also serve in the future for analytical or numerical studies.

Section 3 will then present a simple, analytical model for relating the unsteady rotor forces to the measured wall pressures. This model uses the infinite duct Green's function along with the method of images to approximate the finite-length effects. This simple model provides a novel method for understanding the underlying physics and acoustic features of a simple ducted rotor. Section 4 will compare the experimental results to those predicted by this simple analytical model.

2. Description of experiment

A schematic of the experimental setup used is shown in Fig. 1. The circular duct was machined with an inner diameter of 206 mm and a wall thickness of 6.7 mm. The setup was situated on vibration isolation pads and placed in an anechoic chamber with a low frequency limit of 100 Hz. The inlet flow to the duct was conditioned using thin fabric stretch over a coarse wire mesh. A rounded inlet was attached to allow for a thin, attached boundary layer inside the duct. The surface was relatively rough, which tripped the boundary layer near the duct entrance.

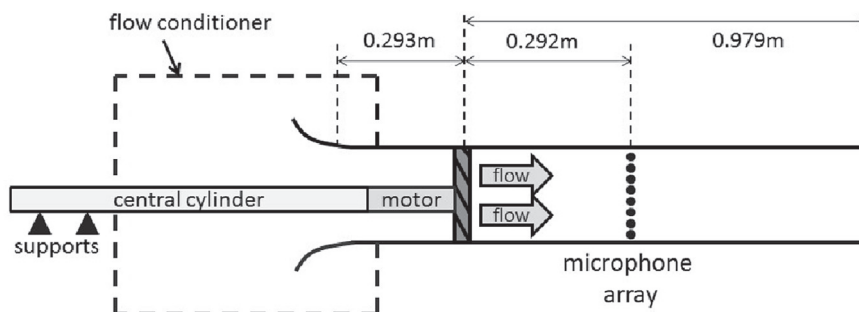


Fig. 1. Schematic of ducted rotor experimental setup.

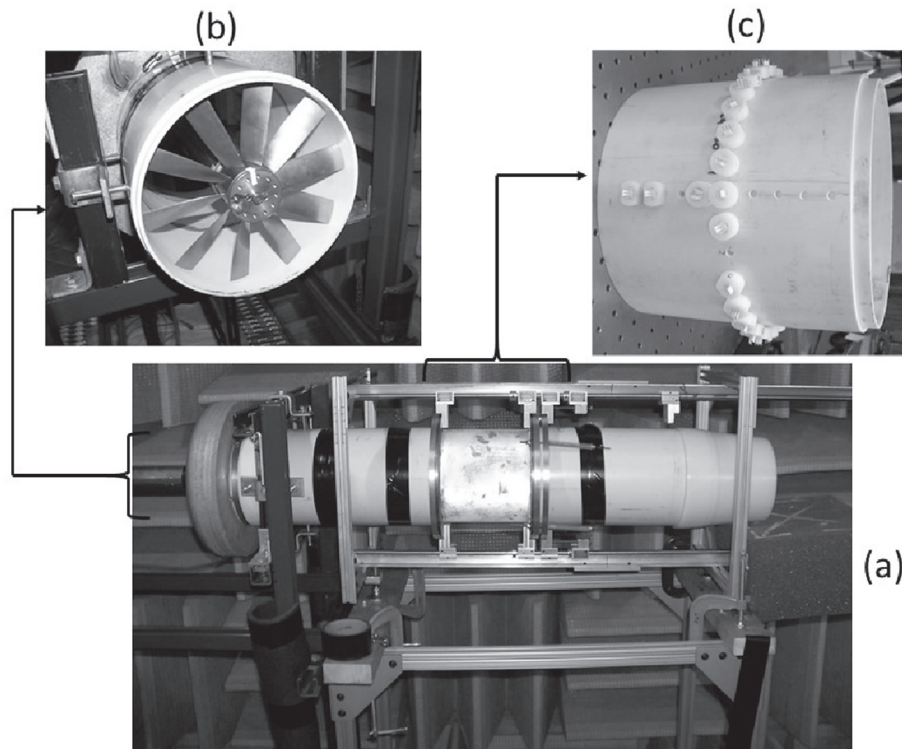


Fig. 2. Pictures of experimental setup. (a) shows the setup as a whole, (b) shows the rotor and central cylinder, and (c) shows the test section for acoustic measurements.

The rotor was supported by a metal cylinder aligned to the axis of the duct. The cylinder was supported by two cantilever supports upstream of the duct as shown. This cylinder housed a compact servo motor which powered the rotor. The rotor was a 10-bladed propeller previously used by Sevik [23]. The rotor had a diameter of 0.203 m and a constant blade chord of 0.025 m. The rotor blades were straight, incorporating twist only with a maximum thickness of 2.4 mm. The rotor operated with a tip clearance that was 5% of the chord. The performance of this rotor was characterized by Stephens and Morris [24], who found a dimensionless flow coefficient of $\phi = 0.3$, and a dimensionless static pressure rise of $\psi = 0.12$ for the rotor conditions used in the present study. ϕ is defined as the approach velocity of the air divided by the rotor tip speed (U_{tip}). The pressure coefficient is defined as $\psi = \Delta P / \frac{1}{2} \rho_0 U_{tip}^2$ where ρ_0 is the density of the fluid.

The circumferential distribution of unsteady wall pressure was acquired using wall mounted Knowles microphones (model FG-23629-C36). An acoustic measurement test section containing a ring of 32 equally spaced sensors was located 0.292 m downstream of the rotor. The array location was chosen to be more than a diameter downstream of the rotor to avoid strong hydrodynamic near-wake effects and was placed sufficiently far from the duct ends so near-field diffraction would not complicate the acoustic field. All microphones were in situ calibrated using a Bruel and Kjaer microphone, and uncertainties are believed to be less than 1 dB over the frequency range of interest. Data were taken at 51 rotor speeds, from 2500 rpm to 5000 rpm at 50 rpm increments. 16 ensembles were taken at each speed, with each ensemble simultaneously sampling all 32 microphones for 1.6 s with a 40 kHz sampling speed.

Fig. 2 shows three different pictures of the test section. Fig. 2-a shows the test setup as a whole in the anechoic chamber. Fig. 2-b shows a picture of the rotor and central cylinder when located within the duct. In Fig. 2-a, the central portion of the duct is an interchangeable test section based on the testing being performed; Fig. 2-c shows the test section used for acoustic measurements with several of the microphones removed.

The speed-dependence of the radiated sound spectrum for this rotor test setup was extensively studied by Stephens and Morris [21]. Fig. 3 shows the sound spectrum (with 1 Hz frequency bins) measured by an on-axis microphone located approximately 2 m external to the duct exit for a rotor speed of 4000 rpm. While the external radiated sound is not the subject of the present study, it is shown here for completeness to provide a general understanding of the noise generated by the setup. Broad-band sound levels can be observed across a wide frequency range. Also, sharp local maxima in the spectra can be observed. The frequencies of these features can be related to either the blade-passing tones of the rotor, organ-pipe type resonances, or the cut-on duct modes. Note that the blade-passing frequency at this rotor speed is 670 Hz. The characteristics of the blade passing tones and responses due to duct geometry will be examined in greater detail in Sections 4.2 and 4.3.

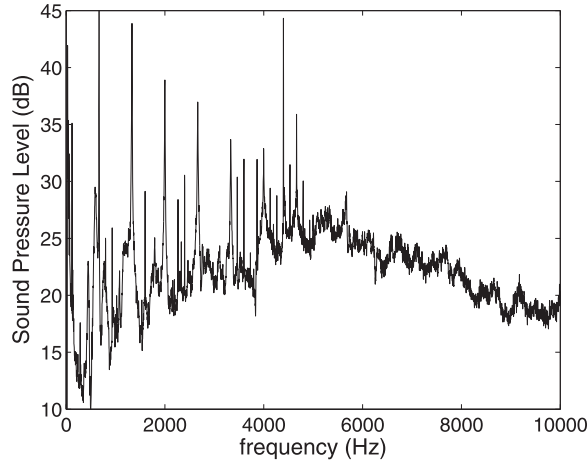


Fig. 3. Farfield sound spectrum with 1 Hz frequency bins for the rotor running at 4000 rpm.

3. Formulation of a rotor acoustic transfer function

This section describes a simple, analytical formulation to relate unsteady rotor forces to the resulting internal acoustic pressure in a finite-length duct. While more sophisticated numerical methods are certainly available, an analytical approach was chosen to provide the best context and understanding of the experimental results described in Section 4.

The sound source was assumed to originate solely from the unsteady lift force resulting from the rotor blades interacting with the casing turbulent boundary layer. Hence, the rotor blades themselves were considered to be represented by acoustically compact dipoles. With this assumption, the acoustic pressure (at a given frequency) can be given by (see Pierce [25]):

$$p(\vec{r}) = \iint \iint_V \vec{F}(\vec{r}_0) \cdot \nabla_0 G(\vec{r}, \vec{r}_0) dV_0 = \iint \iint_V |\vec{F}(\vec{r}_0)| (\hat{n}) \cdot \nabla_0 G(\vec{r}, \vec{r}_0) dV_0. \quad (1)$$

Here, $\vec{F}(\vec{r}_0)$ is a vector representing the dipole force amplitude and direction at source location r_0 . The term $G(\vec{r}, \vec{r}_0)$ is the acoustic Green's function.

3.1. Dipole transfer function

Eq. (1) can be considered as an integration of the product of two scalar quantities by defining the dipole transfer function as

$$T_{dipole} = \hat{n}(\vec{r}_0) \cdot \nabla_0 G(\vec{r}, \vec{r}_0), \quad (2)$$

which allows the pressure to be written as

$$p(\vec{r}) = \iint \iint_V |\vec{F}(\vec{r}_0)| T_{dipole}(\vec{r}, \vec{r}_0) dV_0. \quad (3)$$

The direction of the unsteady forces $\hat{n}(\vec{r}_0)$ was assumed to be normal to the blades. Thus, Eq. (2) can be rewritten as

$$T_{dipole} = \frac{\partial G}{\partial x_0} \cos \varphi + \frac{1}{r_0} \frac{\partial G}{\partial \theta_0} \sin \varphi, \quad (4)$$

where φ corresponds to the blade stagger angle. Fig. 4 illustrates the definition of many of the coordinates associated with a ducted rotor dipole source.

Eq. (4) was evaluated using the Green's function for an infinite, circular duct as presented by Morse and Ingard [26] (Equation 9.2.10 of Section 9.2):

$$G = \frac{e^{ik|x-x_0|}}{2i\pi a^2 k} \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} e^{\frac{J_m(\mu_{mn} \frac{r}{a}) J_m(\mu_{mn} \frac{r_0}{a})}{i\sigma_{mn} \alpha_{mn}}} \cos(m(\theta - \theta_0)) e^{i\alpha_{mn}|x-x_0|} \quad (5)$$

Eq. (5) accounts for both positive and negative values of m , but utilizing the functional symmetry of the Green's function with respect to m , the summation is only performed for non-negative values of m . The Mach number was less than 0.1 for the present experiments and hence is not included in the acoustic propagation.

The derivatives of G in the streamwise and circumferential directions are given by

$$\frac{\partial G}{\partial x_0} = (-1)^b \frac{e^{ik|x-x_0|}}{2\pi a^2} + (-1)^b \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} e^{\frac{J_m(\mu_{mn} \frac{r}{a}) J_m(\mu_{mn} \frac{r_0}{a})}{\sigma_{mn}}} \cos(m(\theta - \theta_0)) e^{i\alpha_{mn}|x-x_0|} \quad (6)$$

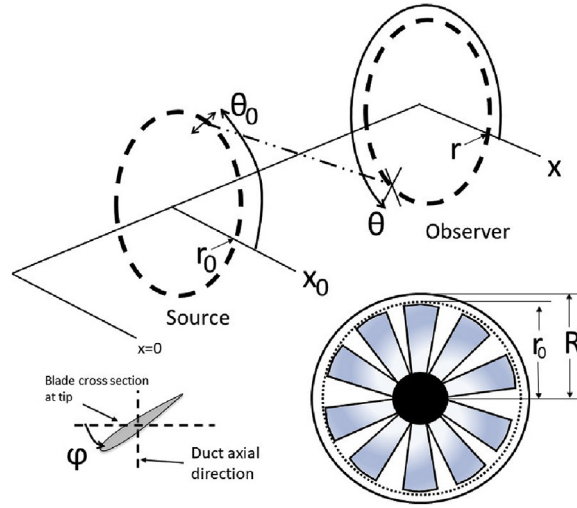


Fig. 4. Illustration of definition of various coordinate nomenclature within duct.

and

$$\frac{\partial G}{\partial \theta_0} = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \epsilon \frac{m J_m(\mu_{mn} \frac{r}{a}) J_m(\mu_{mn} \frac{r_0}{a})}{i \sigma_{mn} \alpha_{mn}} \sin(m(\theta - \theta_0)) e^{i \alpha_{mn} |x - x_0|}, \quad (7)$$

where $k = \omega/c$ is the wavenumber, where ω is the angular frequency and c is the speed of sound in the fluid medium. a is the radius of the duct. For values of $x \geq x_0$ let $b = 0$, and otherwise let $b = 1$. The terms m and n are integer values corresponding to the circumferential and radial mode numbers, respectively. The variable ϵ is equal to $\frac{1}{2}$ for $m = 0$ and equal to 1 otherwise. The term J_m is the Bessel function of the first kind, and μ_{mn} is a zero of the Bessel function's derivative, defined by $J'_m(\mu_{mn}) = 0$. The term α_{mn} is the mode and frequency dependent axial wave number, defined as

$$\alpha_{mn} = \sqrt{\left(\frac{\omega}{c}\right)^2 - \left(\frac{\mu_{mn}}{a}\right)^2}. \quad (8)$$

The normalizing factor σ_{mn} is a mode-dependent value based on Lommel's integral which determines the magnitude of the transfer function and is given by

$$\sigma_{mn} = \pi a^2 \left(1 - \frac{m^2}{\mu_{mn}^2}\right) J_m^2(\mu_{mn}). \quad (9)$$

In Eqs. (6) and (7), x , r , and θ are the axial, radial, and circumferential locations of the observer location, while x_0 , r_0 , and θ_0 are the axial, radial, and circumferential locations of the source location.

To calculate the acoustic pressure, the summations of Eqs. (6) and (7) normally would be taken for all integer values of m from $-\infty$ to ∞ . However, the functions within these summations are even with respect to m , allowing these functions to be calculated once for $m \geq 0$ and then doubled; the exception is the case of $m = 0$, which ϵ accounts for. Substituting the axial and tangential components into Eq. (4) yields

$$T_{dipole}(x, x_0, r, r_0, \theta, \theta_0) = \frac{e^{ik|x-x_0|}}{2\pi a^2} + \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \chi(m, n, x, x_0, r, r_0) \xi(m, r_0, \theta, \theta_0), \quad (10)$$

where

$$\chi(m, n, x, x_0, r, r_0) = \epsilon \frac{J_m(\mu_{mn} \frac{r}{a}) J_m(\mu_{mn} \frac{r_0}{a})}{\sigma_{mn}} e^{i \alpha_{mn} |x - x_0|} \quad (11)$$

and

$$\xi(m, r_0, \theta, \theta_0) = \xi_{ax}(m, \theta, \theta_0) + \xi_{tan}(m, r_0, \theta, \theta_0). \quad (12)$$

Here,

$$\xi_{ax}(m, \theta, \theta_0) = \cos(m(\theta - \theta_0)) \cos \varphi \quad (13)$$

and

$$\xi_{\tan}(m, r_0, \theta, \theta_0) = \frac{1}{ir_0\alpha_{mn}} \sin(m(\theta - \theta_0)) \sin \varphi. \quad (14)$$

The term outside the double summation of Eq. (10) represents the plane-wave, corresponding to the $\langle m, n \rangle = \langle 0, 0 \rangle$ case, while χ includes terms which are not dependent on the blade stagger angle φ or the tangential terms θ or θ_0 . The terms ξ_{ax} and $\xi_{\tan}$ are the axial and tangential contributions to ξ , respectively.

Excluding the plane wave term, α_{mn} is the only frequency-dependent term in the evaluation Eq. (10). The “cut-on” frequency of a mode $\langle m, n \rangle$ is the frequency for which $\alpha_{mn} = 0$ and in general is a unique value for each mode. For a single $\langle m, n \rangle$ mode, the magnitude of the axial component of the dipole transfer function appears similar to a step function with the step occurring at the cut-on frequency. The tangential component of the dipole transfer function is negligibly small at frequencies below the cut-on frequency, approaches infinity at the cut-on frequency, and asymptotically decays for increasing frequency above the cut-on frequency. The phase of the dipole transfer function is dependent on α_{mn} due to the final term of Eq. (11).

3.2. Rotor acoustic source model

The rotor acoustic source was modeled as a “dipole ring” located at 95% of the rotor blade span. This assumes that the unsteady lift generated by the blades is confined to the rotor tip region, because the inflow turbulence is confined to the duct boundary layer. The acoustic pressure generated by a single dipole ring at a single frequency is found by integrating Eq. (3) around the circumference of the dipole ring. The total acoustic pressure is calculated by summing this result for every dipole ring mode M , given by

$$p(x, r, \theta, \omega) = \sum_{M=-\infty}^{\infty} \int_0^{2\pi} \frac{F_M(\omega)}{2\pi} e^{i(M\theta_0 + \psi_M(\omega))} T_{\text{dipole}}(x, x_0, r, r_0, \theta, \theta_0) d\theta_0, \quad (15)$$

where the dipole force per unit angle has been substituted into the integral. Here, M is the circumferential mode of the dipole ring, $F_M(\omega)$ is a real, non-negative value corresponding to the amplitude of the dipole ring for circumferential mode M , and $\psi_M(\omega)$ is an initially-unknown offset of the phase of the dipole ring based on the reference location or time. For the case where $M = 0$ all oscillating forces are in phase, which corresponds to the case described by Ref. [22] where the far-field acoustics can be approximated by a single dipole on the axis of the rotor with an oscillating force of magnitude $F_M(\omega)$.

Assuming that T_{dipole} takes the analytical form detailed in Section 3.1, then the only unknown quantities in Eq. (15) are F_M and $\psi_M(\omega)$, which are the amplitude and phase of the blade lift forcing function for mode M , respectively. Using a similar method as was used on Eq. (3), because $F_M(\omega)$ is independent of θ_0 , Eq. (15) can be recast into a forcing function and transfer function as

$$p(x, r, \theta, \omega) = \sum_{m=-\infty}^{\infty} F_M(\omega) T'_{\text{rotor}}, \quad (16)$$

where

$$T'_{\text{rotor}} = \int_0^{2\pi} \frac{1}{2\pi} e^{i(m\theta_0 + \psi_m(\omega))} T_{\text{dipole}}(x, x_0, r, r_0, \theta, \theta_0) d\theta_0. \quad (17)$$

Two simplifications shall be utilized for evaluation of Eq. (16). The first simplification is that, due to the orthogonality between the exponential term in Eq. (17) and the sine/cosine terms in Eq. (10), the integration with respect to θ_0 results in values of zero unless $m = M$. The interpretation of this is that even though a single dipole can excite every circumferential mode, a dipole ring of mode M can only excite an acoustic response of mode M . Thus, the mode of the acoustic response m and the mode of the rotor forcing function M can be used interchangeably.

The second simplification is related to $\psi_m(\omega)$, which sets the phase of each mode of the dipole ring. Due to the orthogonality condition mentioned previously, the value of $\psi_m(\omega)$ for different circumferential modes have no effect on one another. If multiple rotors are present within the system, the difference between values of $\psi_m(\omega)$ for each rotor will affect how the contributions of each rotor sum together. For a single-rotor system, the value of $\psi_m(\omega)$ is based on what instant in time is selected as the origin of the time coordinate; due to the steady-state assumption, this selection is arbitrary. For simplicity, the time coordinate will be assumed to be selected such that $\psi_m(\omega) = 0$ for all m and ω . Values of the rotor transfer function for which $\psi_m(\omega) = 0$ shall be labeled as T_{rotor} .

The modal circumferential pressure P_m can be determined by performing a Fourier transform with respect to θ of both sides of Eq. (16). Note that the integral in Eq. (17) is with respect to θ_0 , representing an integration around the dipole ring, whereas the Fourier transform is with respect to θ , representing an integration around the circumference of the observer location. Using the two simplifications mentioned previously, application of a Fourier transform to Eq. (16) results in

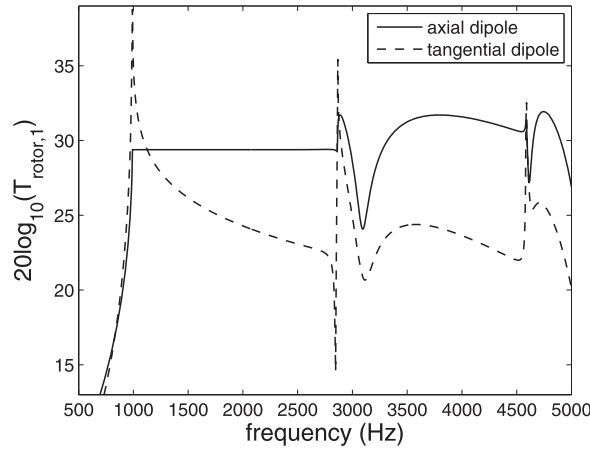


Fig. 5. Axial and tangential components of the acoustic rotor transfer function for $m = 1$. This transfer function assumes a duct of the dimensions utilized in the experiment detailed in Section 2.

$$P_m(x, r, \theta) = F_m(\omega) T_{rotor,m}, \quad (18)$$

where $P_m(x, r, \theta)$ is the coefficient of the Fourier transform for the pressure for mode m , and the final form of the rotor transfer function is given by

$$T_{rotor,m} = \sum_{n=1}^{\infty} e^{\frac{J_m(\mu_{mn} \frac{r}{a}) J_m(\mu_{mn} \frac{r_0}{a})}{\sigma_{mn}}} \pi e^{i\alpha_{mn}|x-x_0|} \left((-1)^b r_0 \cos \varphi - \frac{m}{\alpha_{mn}} \sin \varphi \right). \quad (19)$$

For the case $m = 0$, the plane wave term of $\frac{e^{ik|x-x_0|}}{a^2} (-1)^b r_0 \cos \varphi$ must be added outside the summation on the right hand side of Eq. (19). For each $\langle m, n \rangle$ mode, the magnitude of the rotor transfer function remains constant as a function of axial location, but the phase will vary. The pair of terms inside the last set of parentheses of Eq. (19) results from Eqs. (13) and (14) and determines the relative contributions of the axial and tangential components.

The magnitude of $T_{rotor,m}$ obtained from Eq. (19) was evaluated for the duct, rotor, and microphone geometry described in Section 2. The axial and tangential contributions to $T_{rotor,m}$ for circumferential mode $m = 1$ are shown in Fig. 5. Cut-on frequencies can be observed at 989 Hz, 2865 Hz, and 4587 Hz, corresponding to radial modes $n = 1$, $n = 2$, and $n = 3$, respectively. The “step function like” shape of the axial component and singularity of the tangential component are both observed at the cut-on frequency of the $n = 1$ mode as noted above.

Each α_{mn} changes at a different rate with respect to frequency. Because the phase of each mode $\langle m, n \rangle$ is dependent on $\alpha_{mn}(x - x_0)$, and because $(x - x_0)$ is the same for all modes, the difference in phase between each pair of modes varies with frequency. This creates interference patterns which result in the frequency-dependent fluctuations observed above the $n = 2$ cut-on frequency. At higher frequencies, a greater number of modes are cut on, resulting in more severe frequency-dependent fluctuations.

3.3. Finite duct model

The equations developed in Sections 3.1 and 3.2 were derived for an infinitely long duct. The finite length duct effects will be modeled using a simple method of images. Specifically, the duct ends are considered as an impedance boundary condition that results in a partial reflection of the incident acoustic wave as shown by Levine and Schwinger [27]. The true transfer function of the finite length duct can then be approximated by using the infinite duct model with the addition of “image sources” placed at the appropriate axial distances from the duct end as shown in Fig. 6. The amplitude of the image sources is given by the amplitude of the original rotor forcing function multiplied by the duct end impedance, $R = 0.6$, as suggested by Levine and Schwinger [27]. The condition of $\psi_m = 0$ imposed in Section 3.2 was satisfied with the image rotors. The phase difference between the actual acoustic source and image rotors was fixed to satisfy the boundary conditions at the duct ends.

In principle the method of images suggests an infinite number of image sources should be used to approximate the end effects of the duct. However, the strength of each successive tier of image sources decreases as R^N , where N is the number of image tiers. Because $|R| < 1$, the amplitude will decrease for increasing N . Fig. 7 shows the rotor transfer function for an infinite duct as well as one and three levels of images for $m = 1$.

The interference between the primary acoustic radiation and the reflected sound (modeled with the image sources) creates a complicated function in which narrow-band features are present. These interference patterns are caused mathematically by a

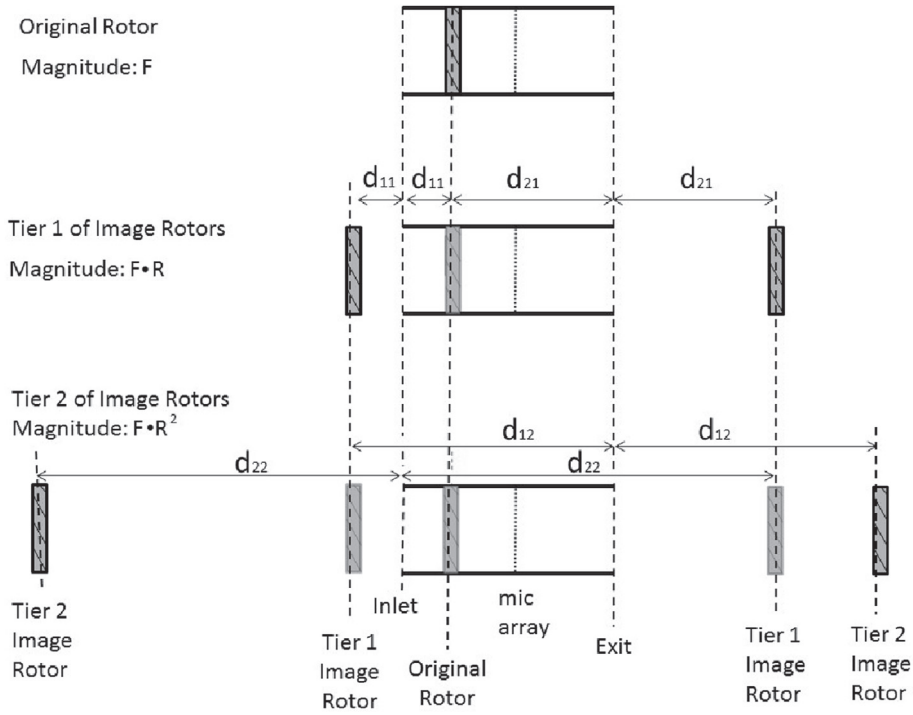


Fig. 6. Example of two tiers of image rotors utilized to satisfy the inlet and exit boundary conditions, with positions approximately to scale. F is the force of the original rotor while R is the reflection coefficient at the inlet and exit to the duct.

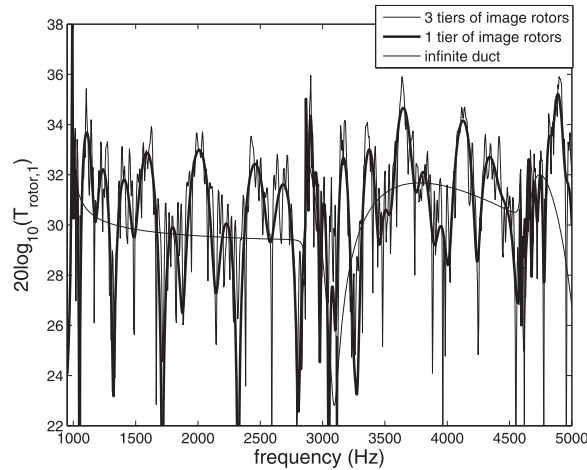


Fig. 7. Theoretical ducted rotor transfer function for no image rotors (i.e. an infinite duct), 1 tier of image rotors, and 3 tiers of image rotors. Transfer function is for circumferential mode $m = 1$.

similar mechanism to that which caused the interference patterns for multiple radial cut-ons. However, in the case of the image rotors, it is the axial separation which varies between the true rotor and image rotors while the value of α_{mn} remains constant, allowing interference patterns to occur for a single radial cut-on. Each successive tier of image rotors will create interference patterns which are smaller in amplitude (due to the R^N dependent amplitudes) and narrower in frequency (due to the increasing $|x - x_0|$).

4. Experimental wall pressure analysis

This section will describe the results obtained from the experiments described in Section 2. Section 4.1 first describes the basic wall pressure measurement results. Section 4.2 then compares the measured forcing function to the theoretical predictions

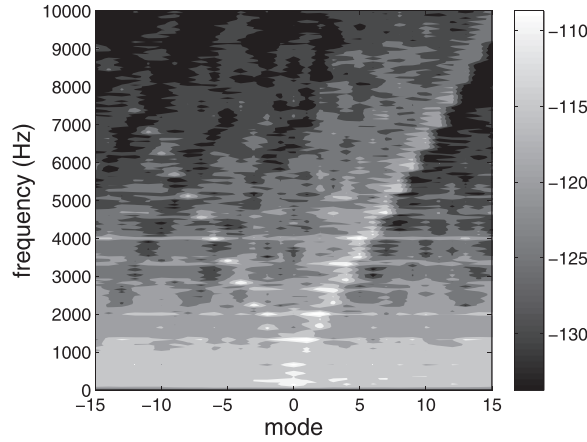


Fig. 8. Auto-spectral density of wall pressure for a rotor speed of 4000 rpm as a function of frequency and circumferential mode number. Color axis equals $10\log_{10}(G_{pp,m}^*)$.

by Stephens and Morris [22]. Section 4.3 compares the results of the measured transfer function to the theoretical rotor transfer function derived in Section 3. Lastly, Section 4.4 examines the experimental spectra as a function of α_{11} , demonstrating a novel way of normalizing the interference patterns were analytically discussed in Section 3.3.

4.1. Wall pressure measurements

Wall pressure data were obtained simultaneously at 32 circumferentially spaced locations for 51 different rotor speeds for the experimental setup detailed in Section 2. The data were initially processed by computing the spatial (circumferential) discrete Fourier transform of the instantaneous wall pressure. The auto-spectral density of the modal amplitudes, $\Phi_{pp,m}(f)$, was then computed. Finally, these results were made dimensionless as

$$G_{pp,m}^*(f) \equiv \frac{[\Phi_{pp,m}(f)]}{(U_{tip}/D)(\frac{1}{2}\rho_0 U_{tip}^2)^2}, \quad (20)$$

where ρ_0 is the density of the fluid, U_{tip} is the tangential velocity at the rotor tip, and D is the duct diameter.

Fig. 8 shows $G_{pp,m}^*$ at a rotor speed of 4000 rpm as a function of circumferential mode m and frequency. The cut-on frequencies at each mode cause a distinctive “V” shape where the high-amplitudes were observed.

There are notable amplitudes of the unsteady surface pressure at frequencies lower than the cut-on frequencies (i.e. “below the V”). These amplitudes are a result of the turbulent rotor wake and the duct boundary layer. These pressure values are predominantly below 1000 Hz, and appear to have little dependence on the mode number, m .

Analysis of the acoustic pressure requires the removal of the hydrodynamic contribution to the pressure. Here we assume that the pressure is a summation of the acoustic and hydrodynamic contributions. The hydrodynamic pressure was found from the data presented in Fig. 8 by assuming that the pressure for a given circumferential mode was entirely hydrodynamic for $f < f_{cut-on}$. The auto-spectral density of several of the higher modes are shown in Fig. 9. These results show little dependence on the circumferential mode as noted above. Furthermore, the results were found to scale well with rotor speed using Eq. (20). Due to these factors, the dimensionless hydrodynamic pressure spectrum was estimated as a function of frequency by averaging the dimensional spectra for all modes and rotor speeds such that $f < f_{cut-on}$. This hydrodynamic wall pressure spectrum was subtracted from the measured wall pressure at a given rotor speed (when redimensionalized and scaled for the appropriate rotor speed) to obtain the acoustic wall pressure spectrum. Note that all further references to wall pressure in the article will be referring solely to the acoustic wall pressure derived from this method.

The acoustic wall pressure can be considered to be the result of the rotor source amplitude of a given circumferential mode, F_m , multiplied by a transfer function, T_m . Experimentally, microphone measurements can be used to separate and quantify these functions. Methods for doing so, termed “F-T separation algorithms”, generally utilize measurements obtained over a range of rotor speeds in order to distinguish the two functions. The measurements shown here were repeated for a range of rotor speeds from 2500 rpm to 5000 rpm in 50 rpm increments. The algorithm used in this article is based on the algorithm used by Stephens and Morris [21].

The transfer function was assumed to have no dependence on rotor speed, while the magnitude and the frequency of features of the forcing function are assumed to scale with the rotor speed. As a result, as the rotor speed is increased, features of the pressure spectra which increase in frequency are attributed to the forcing function while stationary features are attributed to the transfer function. The F-T separation algorithm is only capable of identifying features in the transfer and forcing functions, but is not capable of independently determining the broadband magnitude of either function. To account for this, the experimental transfer function was given a broadband magnitude equal to that predicted by Eq. (19). This allows the broadband

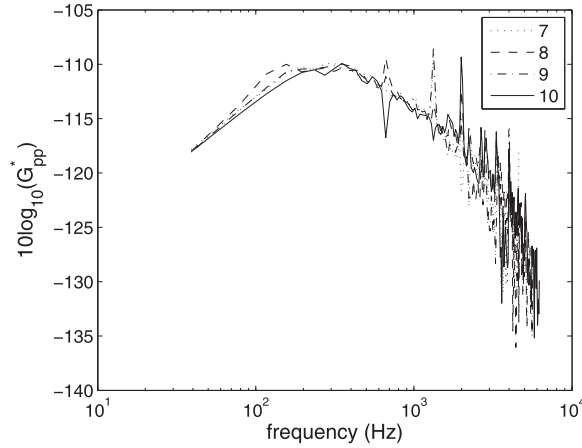


Fig. 9. Wall pressure for various circumferential modes where acoustics are cut off.

magnitude of the experimental forcing function to be determined, but has no effect on the features of the spectra for either function.

In summary, the unsteady surface pressure in the duct was measured by the microphone array and shown as a spectral density function for each circumferential mode (Fig. 8). Values below the acoustic cut-on frequency were used to estimate the hydrodynamic contribution to the spectra (Fig. 9). The surface pressures from the acoustic sources were then found by subtracting the hydrodynamic contribution from the raw spectra. These results, obtained over a wide range of rotor speeds, were used with an F-T separation algorithm in order to separate the effects of the rotor acoustic source amplitude from the net effective transfer function, each of which will be examined in the following sections.

4.2. Experimentally derived forcing function

The experimental forcing functions for modes -3 through 3 are plotted versus frequency normalized by blade passing frequency in Fig. 10. The amplitudes of the spectra are scaled as

$$F^* \equiv \frac{[\Phi_{FF}(f)]_{rotor}}{8\rho_0^2 U_{tip}^3 r_{tip}^5}, \quad (21)$$

where $[\Phi_{FF}(f)]_{rotor}$ is the auto-spectral density of the force and r_{tip} is the radius of the rotor. Note that acoustic results, and hence the rotor forcing function estimates, were only available at frequencies above the cut-on for a given mode. The results are separated into four plots with different mode numbers for clarity.

Fig. 10 also shows the theoretically-derived forcing function predicted through the approach noise model given by Stephens and Morris [22]. In this model the approach turbulence was considered to be independent of the circumferential mode number.

The forcing function for mode 0 shows good agreement with the predictions by Steven and Morris [22]. The magnitudes of the spectra in the regions of integer multiples of the blade passing frequency also agree well, although the theoretical amplitudes were generally higher in amplitude at the local maxima. This was also found by Stephens and Morris [22], and was attributed to approximations made regarding the spatial correlations in the approach turbulent boundary layer. Examination of the other modes shown in Fig. 10 show little dependence of the forcing function on mode number.

4.3. Experimentally derived transfer function

The experimental transfer function derived from the F-T separation algorithm is shown in Fig. 11, which shows the same overall “V” shape as the pressure spectra. In order to better see the details of the transfer function, Fig. 12 and Fig. 13 show the experimental transfer functions for circumferential modes $m = 1$ and $m = 2$, respectively, as a function of frequency. These are plotted alongside the theoretical transfer function for an infinite duct and the predicted transfer function for a finite duct with 3 tiers of image rotors.

The experimental transfer function qualitatively matches the finite duct transfer functions. Both plots show the same types of interference patterns predicted in Section 3.3 from the interactions of the image rotors, and the frequency scales over which these fluctuations occur are similar for the experimental and theoretical transfer functions. Note that the amplitudes of these fluctuations are on the order of 5 dB–10 dB.

Further investigation into the theoretical transfer functions found that the exact frequencies and amplitudes of the interference-induced features can be very sensitive to the duct parameters. As an example, Fig. 14 shows the theoretical rotor

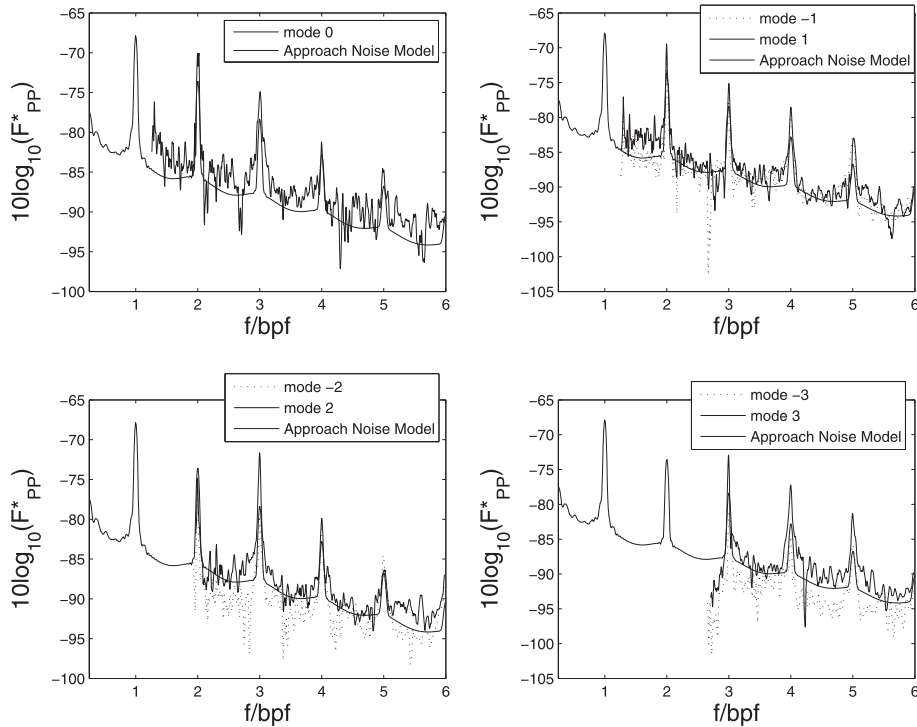


Fig. 10. Rotor forcing at various modes compared to predictions based on approach noise model. Frequency is normalized by blade passing frequency.

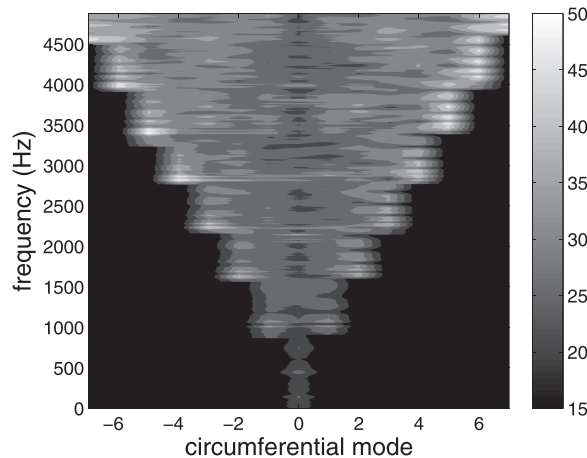


Fig. 11. Experimentally-derived transfer function as a function of frequency and mode. Contours represent $20\log_{10}$ of the transfer function magnitude in Pa/N.

transfer function (for $m = 1$) for the duct detailed in Section 2 and two examples of a longer duct. Only a single tier of image rotor is used in the finite duct transfer functions so that the smaller-scale interference patterns do not make it difficult to visualize the largest-scale interference patterns. In this figure, the length of the duct from the rotor to duct outlet is increased by 2% or 10%, with all other parameters unchanged. The infinite duct transfer function is also plotted to represent a baseline case where no interference patterns are present. One will note that all of the finite duct transfer functions look qualitatively similar. However, the frequencies of the spectral features vary sufficiently between these three functions that, for a fixed frequency, there can be a notable difference between the amplitudes of the three functions. The three functions are the most similar at frequencies slightly above a cut-on frequency and gradually diverge from one another for increasing frequencies. The case of 2% added length diverges more slowly than the case of 10% added length, indicating that these differences are not simply random.

The exact frequencies and magnitudes of the small-frequency-scale peaks were not found to agree between the theoretical and experimental rotor transfer functions. Several reasons for the differences were identified. First, as stated in Section 3, a formal, complete calculation of the acoustic pressure would require the integration of the unsteady lift force over the

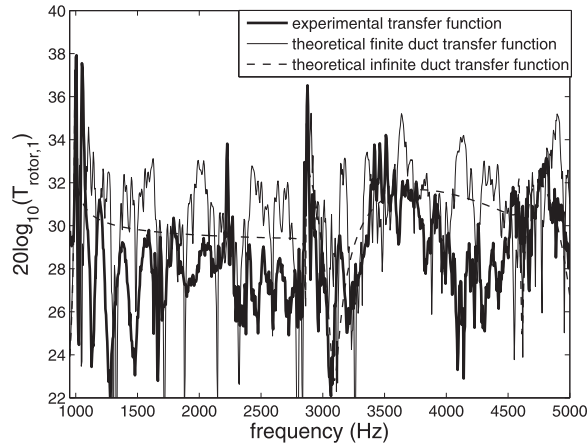


Fig. 12. Comparison of theoretical and experimental rotor transfer function for circumferential mode $m = 1$. $T_{rotor,1}$ is expressed in Pa/N.

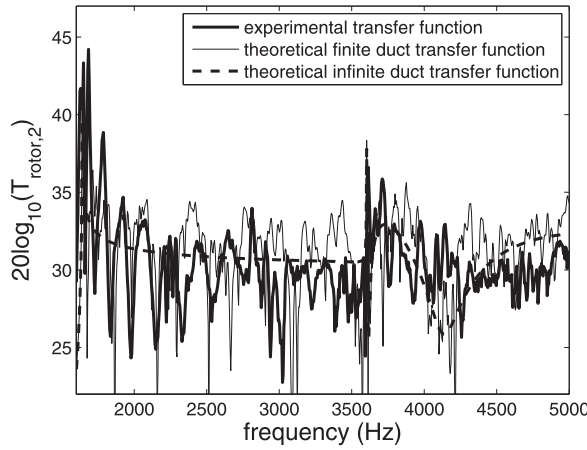


Fig. 13. Comparison of theoretical and experimental rotor transfer function for circumferential mode $m = 2$. $T_{rotor,2}$ is expressed in Pa/N.

entire volume of the duct. In the present case, this integration was approximated by a single ring of lift forces at a single axial location. Second, as demonstrated in Fig. 14, the exact frequencies and magnitudes of these small-frequency-scale peaks are very sensitive to the exact duct parameters. For example, the length of the duct and the position of the microphone array were measured with an accuracy of approximately 1 mm, which is sufficient to vary the frequency values of some of these peaks.

Also, the use of the impedance values from Ref. [27] is considered to be a substantial approximation of the acoustic field. The experiment included an inlet that included a radius of curvature in order prevent boundary layer separation, which likely influenced the amplitude of the end reflections. The authors explored a wide range of impedance values in their study. Very small changes in the impedance values in the analytical prediction resulted in substantial changes in the frequency values of the observed “small scale” spectral features of the acoustic transfer function. Overall agreement with the experiments could only be improved over a relatively narrow frequency range, while reducing the agreement in other ranges. Full implementation of more advanced impedance models could be considered as future work. These observations highlight the limitations of the predictive capability of the simplified formulations presented here. However, the sensitivity of the spectral features to very small changes in the physical parameters (e.g., duct length) or model parameters (e.g., end impedance values) adds insight into the nature of the constructive/destructive patterns that were observed in the experimental results.

4.4. Alpha scaling

The interference patterns in the experimentally-derived pressure spectra and transfer functions could lead an observer to believe these fluctuations are random in nature. However, by examining the relationship to $\alpha_{mn} |x - x_0|$, it can be observed that these variations are actually structured rather than random. Evaluation of Eq. (19) is identical for the real rotor and all image rotors except that (1) each rotor will have a different value of $|x - x_0|$ and (2) each rotor will be scaled by a different

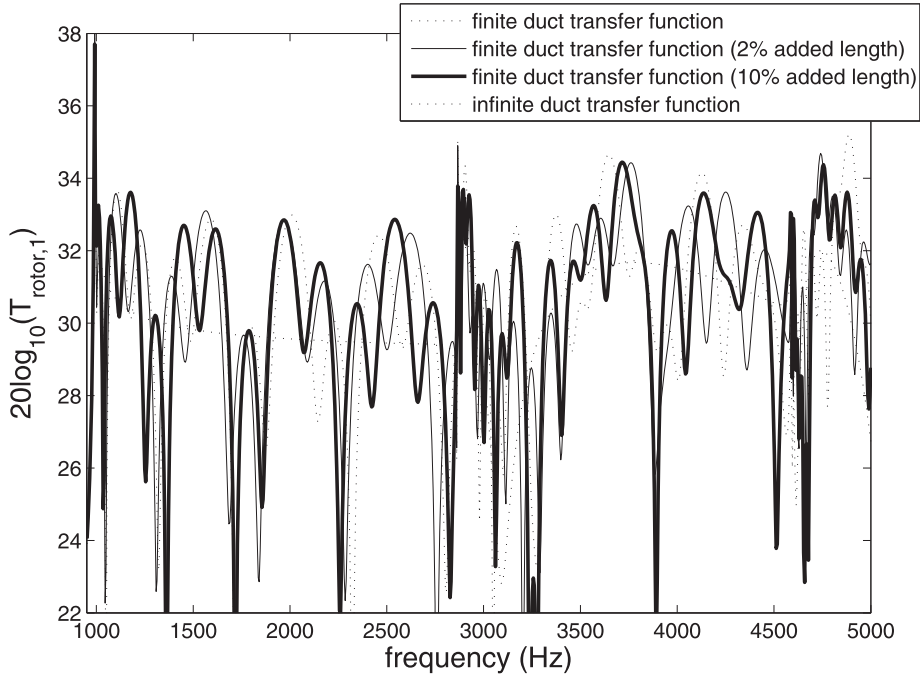


Fig. 14. Comparison of theoretical rotor transfer function for minor variations in duct length, for circumferential mode $m = 1$. $T_{rotor,1}$ is expressed in Pa/N.

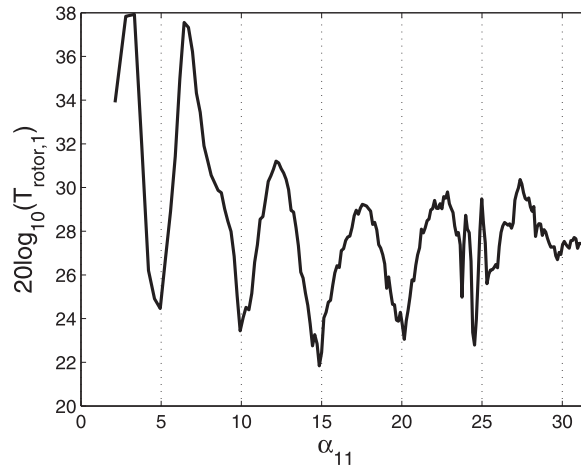


Fig. 15. Experimentally-determined transfer function for circumferential mode $m = 1$ plotted versus α_{11} .

exponentiation of the reflection coefficient based on the tier of the image rotor. Because Eq. (19) is multiplied by a term $e^{i\alpha_{mn}|x-x_0|}$, it is expected that the summation of Eq. (19) across all real and image rotors will result in a function that is repetitive in α_{mn} ; the differing values of $|x - x_0|$ for each rotor will determine the “period” of this function.

Fig. 15 shows the experimental transfer function for mode $\langle 1, 1 \rangle$ using α_{11} as the independent variable. The frequency range used in the figure is from 995 Hz to 2000 Hz, for which only radial mode $n = 1$ is cut on, which means that the only value of α_{mn} which the function will be dependent on is α_{11} . If two radial modes were cut-on, then a portion of the transfer function would scale with α_{m1} and a portion would scale with α_{m2} .

Fig. 15 readily shows that the experimentally-determined function is repetitive in α_{11} . Due to the complexity of the function to calculate α_{mn} (see Eq. (8)), it is highly unlikely that the experimental function would be repetitive in α_{11} only by chance. This provides further justification that the fluctuations are indeed structured (as predicted by the theoretical models presented in this work) rather than being the result of random experimental noise.

5. Conclusions

The purpose of this work was to study the internal acoustics of a ducted rotor using an experiment in which the only acoustic source was the interaction of the rotor tips moving through the duct wall turbulent boundary layer. The measurement results provide insight into the relationships between the source, the duct modes and cut-on effects, and the relatively complicated spectral features created by the finite-duct effects. The simplicity of the geometry allowed a Green's function approach for modeling the relationship between the rotor sources and the duct wall pressure. This model was used to provide context and understanding of the measurements.

The measured unsteady pressure was considered to be a summation of hydrodynamic and acoustic contributions. The fluctuations below the acoustic cut-on frequency were assumed to be purely hydrodynamic, and independent of circumferential mode number. The resulting hydrodynamic pressure spectra was found to have a shape that was similar to simple boundary layer broad-band spectra with the addition of local maxima at multiples of the blade-passing frequency. Hydrodynamic scaling of the pressure spectra (Eq. (20) and Fig. 9) demonstrated the expected collapse.

The acoustic component of the measured pressure spectra was considered to be a product of a rotor source function and an acoustic transfer function. The resulting source functions were found to be independent of circumferential mode order, as is expected given the relatively small scale turbulence of the duct boundary layer. The amplitude of the measured source spectra was found to agree with theoretical predictions to within a few dB over much of the frequency range.

The measured acoustic transfer function was found to exhibit very complicated features with large local maxima and minima. The “size” of these features in the frequency domain varied from tens to hundreds of Hz, with amplitudes varying from a few dB to as much as 15 dB. The analytical model developed from the duct Green's function showed similar features, and demonstrated that these features are a result of the finite length duct effects. The simple formulation provided a method to predict the overall amplitudes, trends, and features of the acoustics in a simple, finite-length duct. However, the details of the acoustic transfer function could only be predicted qualitatively. Attempts to “tune” the model with varied duct parameters (e.g., effective lengths and reflection coefficients) would provide agreement at some frequencies but not at others.

Author contributions

The following contributions were made to the present work by the authors.

Jason Tomko: Acquired data published in manuscript. Derived equations and performed calculations in manuscript. Primary author of manuscript.

Scott Morris: Primary advisor to all work done. Provided significant contributions and review to writing of manuscript.

David Stephens: Performed the initial data runs and processing. Wrote the force – transfer function separation script utilized in the data processing.

Tom Economon: Designed and built test setup used in the current work. Assisted in collection of data.

Acknowledgments

The authors of this document would like to acknowledge Dr. Hafiz Atassi for contributing simplified equations of the acoustic transfer function for a monopole within a duct, which was utilized in the derivation of the rotor transfer function.

The authors would like to thank the Office of Naval Research for their support of this work, D. Nalchajian, program manager, program number N000014-12-1-0337. The first author would also like to thank Richard and Peggy Notebaert for their support given through their premier fellowship at the University of Notre Dame.

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